CGMYW Lévy Model: P-then-Q Calibration on AAPL

Progress Report

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1 Summary

I built and tested the full two-stage pipeline for the 8-parameter asymmetric CGMYW model $(C_n, C_p, G, M, Y_n, Y_p, W_n, W_p)$, using AAPL:

1. **P-measure:** fit to daily CRSP returns over a trailing 252-trading-day window by Anderson–Darling-weighted tail-probability matching (30 thresholds spanning the distribution, 8-start optimizer).

2. **Q-measure:** Carr–Madan FFT pricer; calibrate to a 30–45 day OTM option smile using a vega-weighted price error. M is solved from the martingale condition $T\,\psi_Q(-i) = \ln(F/S_0)$, so every candidate is a martingale measure. Main fit (per our 2026-10-19 discussion): C_n, C_p, G, W_n, W_p free, Y_n, Y_p pinned at P values.

Headline: the main Q fit matches the market smile to **0.97** and **0.74** implied-vol points on the two test dates. The automated test suite passes in full (**138/138 tests**).

New (2026-11-02): a 9-parameter Sato CGMYW fit to four maturities at once (1 month to 1 year) has a median error of **0.99 vol points** over 18 dates and beats the Lévy model on every date; W adds little to the surface fit (Section 6).

2 Correctness checks

Each piece is checked against an independent reference before any market data is used.

Check	Reference	Result
Closed form with $W_n = W_p = 0$, $C_n = C_p$, $Y_n = Y_p$	classical CGMY ψ	error $< 10^{-8}$ over $\lvert u\rvert \le 300$
Power-tilt term ($W \neq 0$)	direct 30-digit quadrature of $\int (e^{iux} - 1)k(x)dx$	rel. error $\sim 10^{-14}$
Kummer U for complex z	mpmath	passes (both branches)
Lévy density $\to$ PDF normalization	integrates to 1	passes
Carr–Madan FFT	Black–Scholes closed form	error $< 10^{-5}$ (Fig. 1)
CGMYW option prices	Gil–Pelaez quadrature	error $< 10^{-4}$
Risk-neutralization	Madan–Wang Table 2, all 10 tickers	G, M within 0.2%; W within 0.22
Synthetic smile	known Q parameters	recovered, IV RMSE < 0.2 vol pts

On the Table 2 reproduction: started from the paper's own values or from the P values, our minimum-norm solve lands on the same point. That point meets the martingale condition to 10^{-12}; the published values meet it only to about 10^{-5}. The small gap in W therefore comes from the paper's solver tolerance, not from a model error.

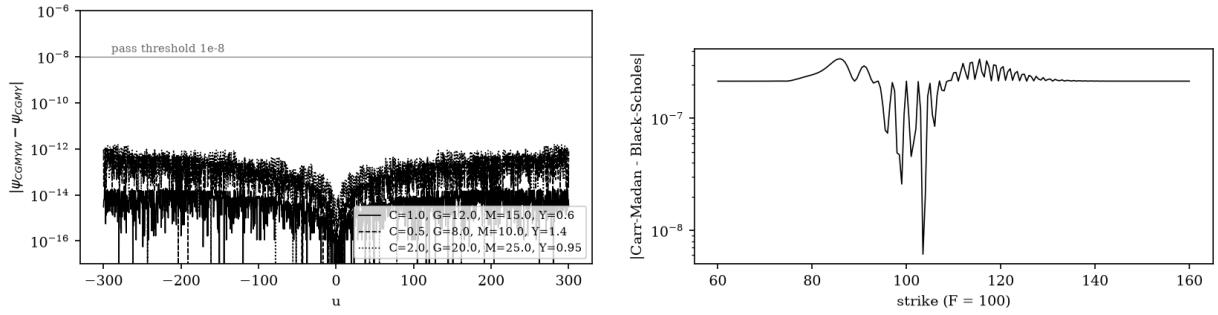


Figure 1: Left: CGMYW reduces to classical CGMY (error at machine precision). Right: FFT pricer vs. Black–Scholes.

3 P-measure results (AAPL)

Window end	C_n	C_p	G	M	Y_n	Y_p	W_n	W_p	AD obj.
2023-01-13	3.803	3.920	118.2	143.9	0.053	0.061	14.86	4.83	0.0151
2025-08-29	0.543	0.888	41.0	94.5	0.043	0.051	14.57	5.03	0.0206
Madan–Wang T1	0.952	1.344	52.0	97.6	0.003	0.003	22.27	4.83	–

Table 1: Daily units; 251 returns per window; all 8 optimizer starts converged on both dates. The paper uses a fixed 1033-day window, but our fits land in the same regime: small Y, $W_n \gg W_p$, $M > G$.

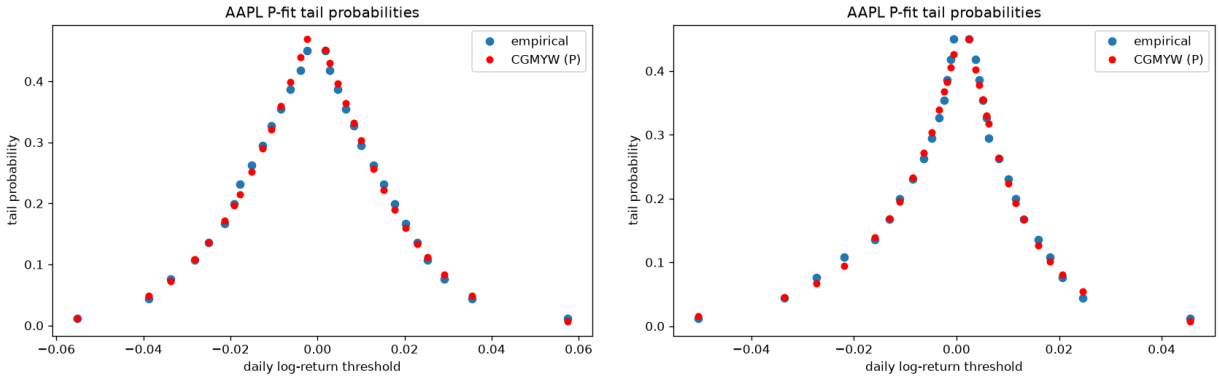


Figure 2: Empirical vs. model tail probabilities. Left: 2023-01-13. Right: 2025-08-29.

4 Q-measure results (AAPL)

IV RMSE (vol points)	2023-01-13 → 02-24 (49 opts)	2025-08-29 → 10-10 (24 opts)
P + martingale only (no option data)	6.24	7.71
Strictly equivalent (C, Y pinned)	6.29	8.99
Main: C_n, C_p free, Y pinned	**0.97**	**0.74**
Unrestricted (all 8 free, diagnostic)	0.42	0.14

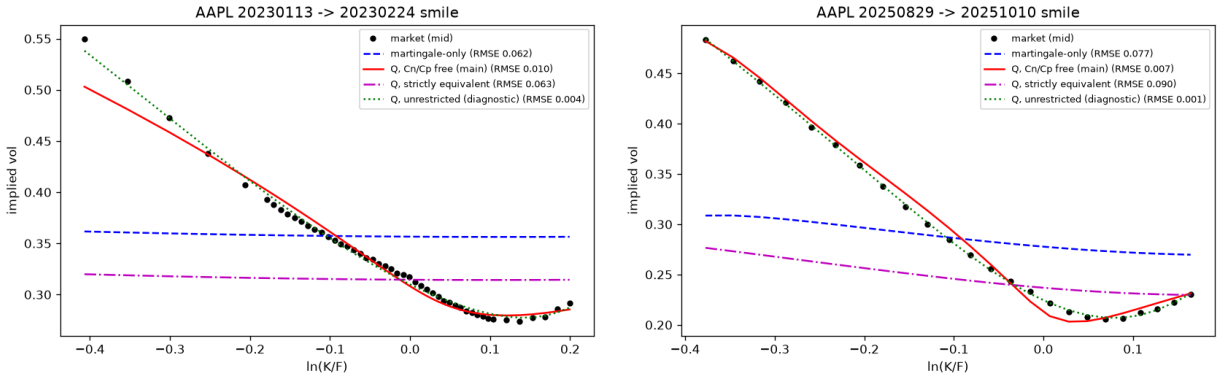


Figure 3: Market vs. model implied-vol smiles, 42-day maturity.

Main-fit parameters (annual): 2023-01-13: $C_n = 9.84, C_p = 26.98, G = 8.80, M = 29.23, W_n = 2.61, W_p = 5.67$. 2025-08-29: $C_n = 4.29, C_p = 11.07, G = 0.035, M = 27.28, W_n = 11.0, W_p \approx 0$. On the second date the down tail is carried almost entirely by the power tilt W_n, which is the role the model was designed for.

5 Stability across 18 quote dates

The full pipeline (fresh 252-day P-fit, then the main Q fit) was rerun on 18 quote dates, about every two months from 2023-01 to 2025-08. Out of sample: the fitted Q parameters were used, without refitting, to price a longer 60–120 day smile on the same date.

	Median	Range	
Main Q fit, IV RMSE (vol pts)	1.18	0.44–5.38	under 2 on 14 of 18 dates
Out-of-sample maturity, IV RMSE (vol pts)	2.83	0.76–6.54	no refit
P tail-probability RMSE	0.0107	0.0063–0.0164	Madan–Wang AAPL: 0.0067 (1033 days)

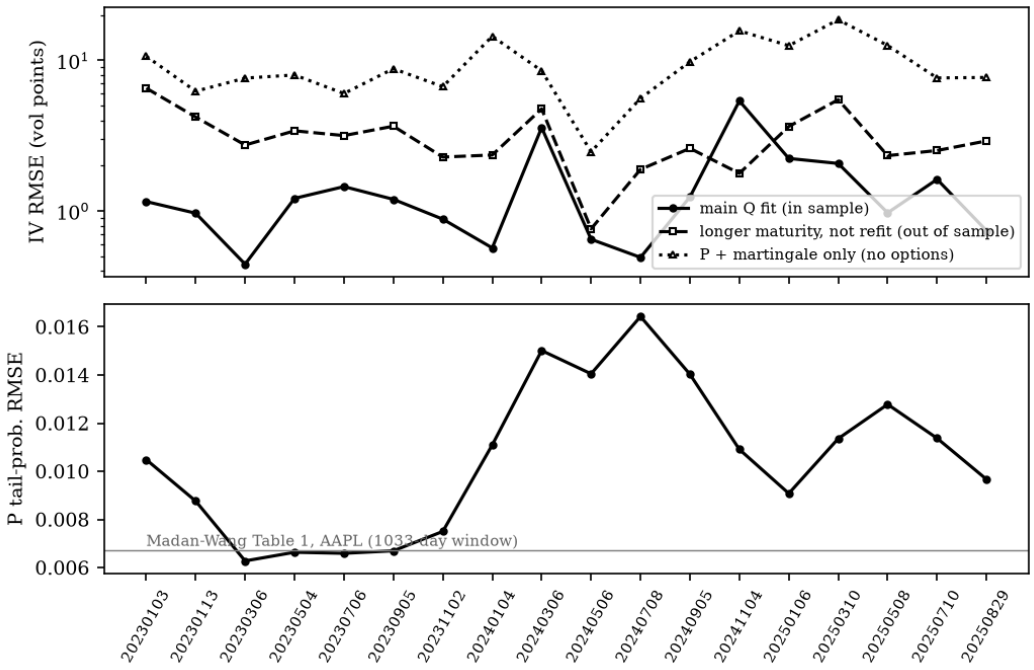


Figure 4: Top: Q-fit error over time (log scale): in sample, out of sample, and P + martingale only. Bottom: P-fit tail-probability RMSE per window.

The two weakest in-sample dates are 2024-11-04 (5.38) and 2024-03-06 (3.58). On every date, the option-free martingale-only measure is far worse than the calibrated fit, so the option data is doing real work. The out-of-sample error is roughly 2–3× the in-sample error: the parameters carry over across maturities, but not perfectly.

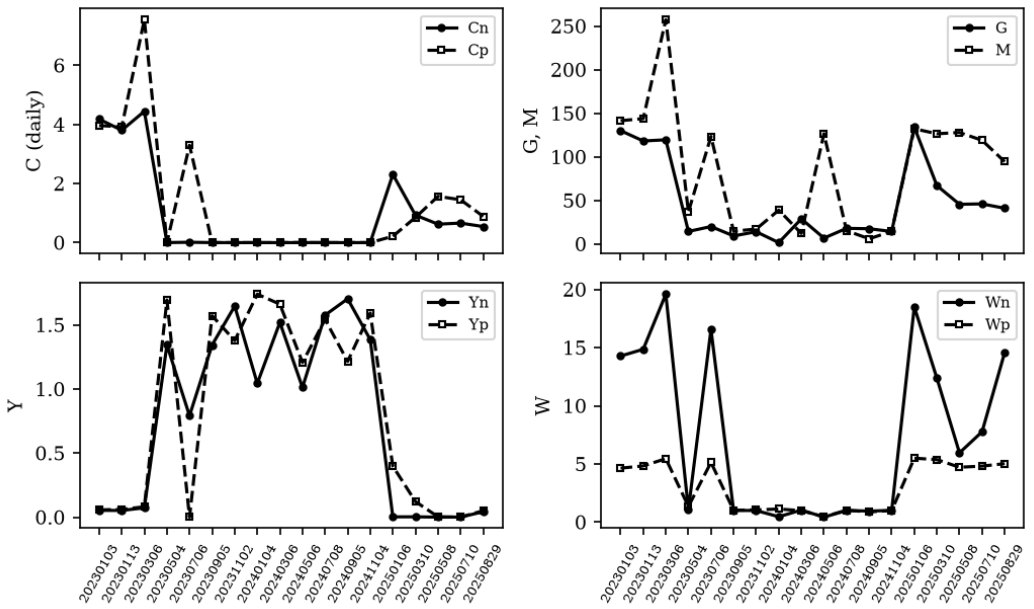


Figure 5: P-measure parameters across the 18 trailing windows.

6 Option surface: Lévy vs. Sato, with and without W

Following the 2026-11-02 meeting, a second calibration fits the whole option surface directly.

- **All parameters free.** The martingale condition is met by a constant, $\ln S_T = \ln F + X_T + \omega_T$ with $\omega_T = -\ln E[e^{X_T}]$, instead of by solving for M.
- **Forward from the options.** $F = S_0 e^{(r-q)T}$, with q implied so that F is where call and put mids cross (put-call parity). It lands within 0.5% of WRDS's forward; implied $q \approx 0.7$–1.3%.
- **Four maturities at once:** the traded expiries closest to 1, 3, 6 and 12 months, one parameter set.
- **Sato process.** $X_T = T^\gamma X_1$ with X_1 CGMYW, so $E[e^{iuX_T}] = e^{\psi(uT^\gamma)}$: 9 parameters. CGMYW is self-decomposable ($|x|k(x)$ decreasing in $|x|$ for $G, M, Y, W \geq 0$), so this is a valid Sato process. Under a Lévy process, skew decays like $T^{-1/2}$ and excess kurtosis like T^{-1}; under Sato both are flat in T.

Four models on the same 18 dates with the same objective: Lévy CGMY (6 parameters), Lévy CGMYW (8), Sato CGMY (7), Sato CGMYW (9). Each CGMYW fit is warm-started from its CGMY fit, since it contains CGMY as the case $W = 0$.

	Surface IV RMSE (vol pts)		Mean IV RMSE by maturity			
Model	median	range	∼1m	∼3m	∼6m	∼1y
Lévy CGMY	1.56	1.11–4.22	2.86	1.25	1.02	1.68
Lévy CGMYW	1.52	1.07–4.22	2.81	1.21	1.01	1.63
Sato CGMY	1.00	0.43–1.52	1.86	1.00	0.47	0.89
Sato CGMYW	**0.99**	0.43–1.52	1.85	0.98	0.45	0.88

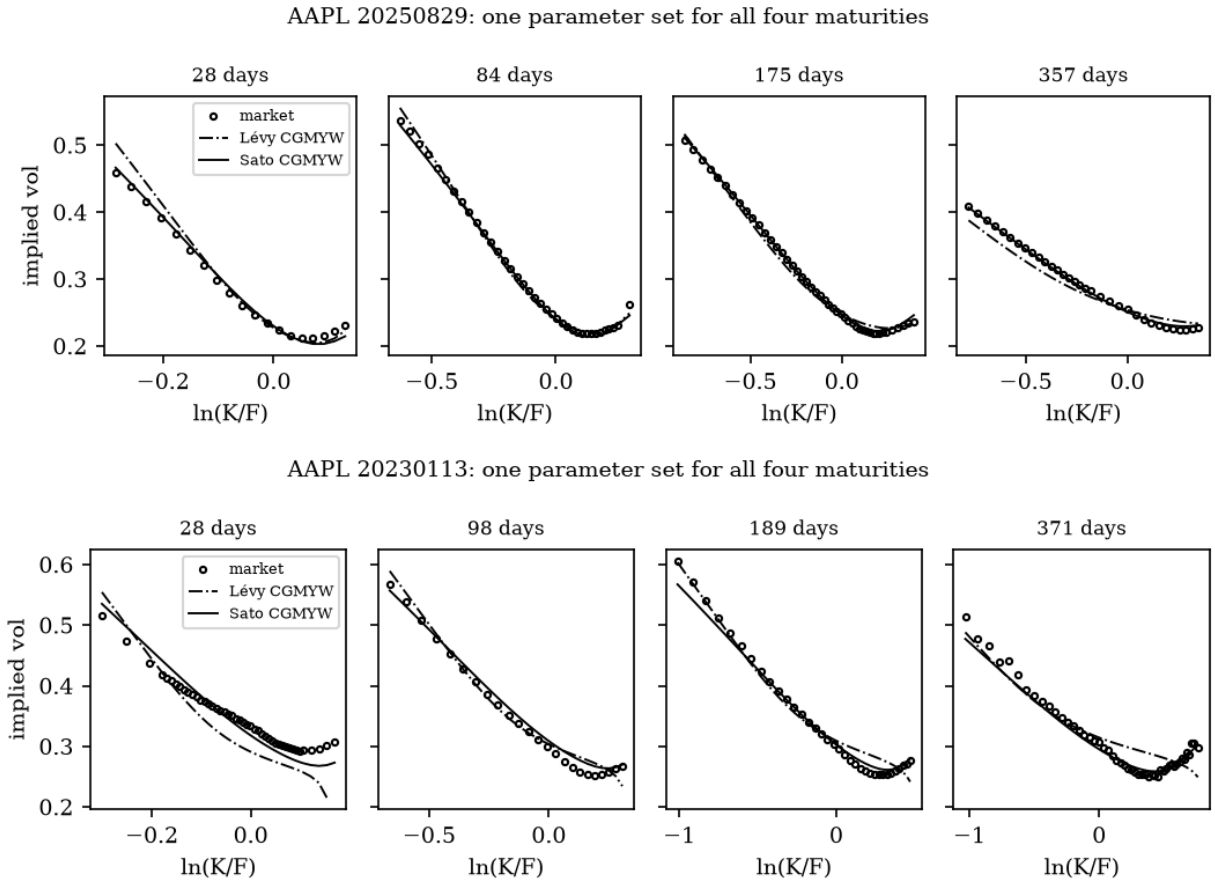


Figure 6: One parameter set for four maturities. Lévy (dash-dot) is too steep at 1 month and too flat at 1 year; Sato (solid) follows the term structure.

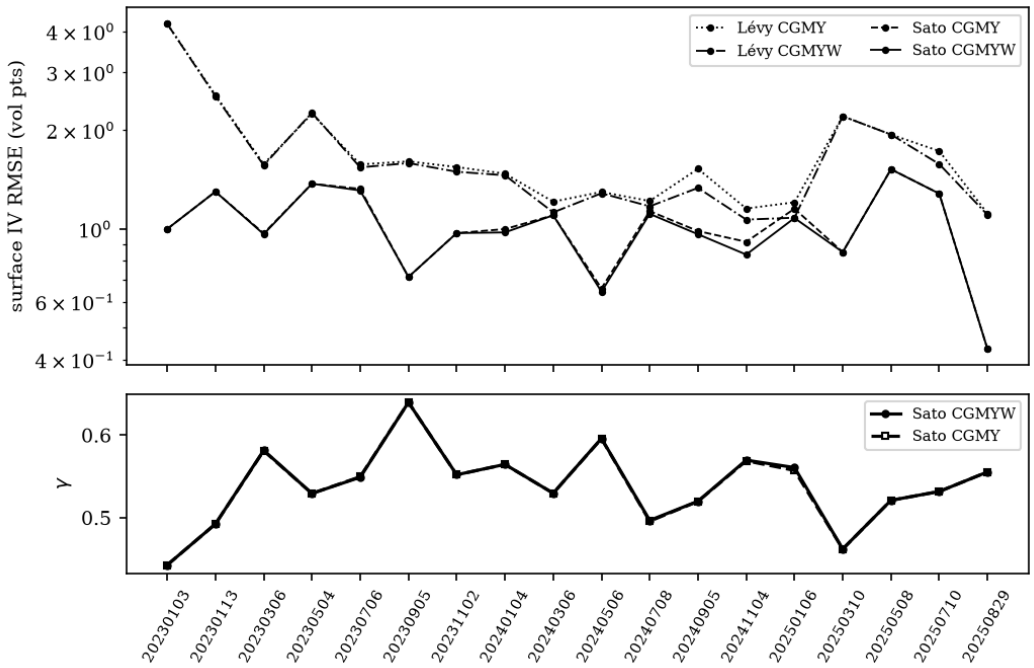


Figure 7: Top: surface IV RMSE per date for each model (log scale). Bottom: fitted γ.

Results.

- **Sato beats Lévy on all 18 dates**, with a median surface error of 1.0 vs. 1.5 vol points. Lévy fails at the ends of the term structure (2.9 vol points at 1 month), as the $T^{-1/2}$, T^{-1} argument predicts.
- γ **is stable:** median 0.54, range 0.44–0.64.
- W **adds little on the option surface.** Under Sato, the fit uses W on only 6 of 18 dates (W_n from 0.6 to 2.2, $W_p \approx 0$ throughout), and the error falls by at most 8% (2024-11-04: 0.91 $\to$ 0.84); the median gain is 0. Under Lévy, W lowers the fit objective on 15 of 18 dates; the IV error falls by a median 1.2% and at most about 12% (2024-09-05: 1.52 $\to$ 1.33). Where W is used, it is always on the down side, as in the P fits ($W_n \gg W_p$).
- **Caveat:** Y_p often sits at its lower bound and $Y_n \approx 1.3$–1.7, which points to weak identification of the up tail from four smiles. Nelder–Mead is a local method, but the nested consistency (CGMYW never worse than CGMY) now holds on every date.

7 Findings

- The pricer and optimizer are not the bottleneck: the unrestricted fit reaches 0.14–0.42 vol points.
- A strictly equivalent change of measure (C, Y fixed) cannot produce AAPL's put skew (6–9 vol pts). With no diffusion term, the martingale condition forces the two tails to nearly cancel.

- Freeing C_n, C_p (as in the Markovian-economies paper) closes most of the gap: under 1 vol point. Risk-neutral C comes out below physical C on both sides.
- (C, Y) are only weakly identified from 251 daily returns, and those are exactly the parameters pinned in Q.
- Across maturities, the time dimension matters far more than W: Sato vs. Lévy is worth about 0.5 vol points; W is worth about 0.02 under Sato.

8 Next steps

Run the Lévy/Sato, W/no-W comparison on all 10 stocks (no stock-specific code paths exist); check whether W matters more for names with heavier down tails; tighten identification of the up tail (Y_p at its bound); build the estimation notebook.